

A disproof of Conjecture 00000003963

The algebraic connectivity of a hypercube layer is not $\Theta(\log n/n^2)$

1 The conjecture

Conjecture 00000003963 reads:

Definition: A cube layer is the set of vertices of fixed weight in the hypercube. *Conjecture:* The algebraic connectivity of the induced graph of a layer, a Johnson graph, attains its minimum over all layers at $l = n/2$, of order $\Theta(\log n/n^2)$.

The Chinese version says the same: the algebraic connectivity of the induced graph (Johnson graph) of layer ℓ is smallest at $\ell = n/2$ among all layers, and its order is $\Theta(\log n/n^2)$.

We identify a vertex of the hypercube $Q_n = \{0, 1\}^n$ with the subset of $[n] = \{0, \dots, n-1\}$ where it equals 1; its weight is the size of the subset. The layer of weight ℓ is the set of ℓ -subsets of $[n]$.

Definitions used.

- *Hypercube* Q_n : subsets $s, t \subseteq [n]$ are adjacent iff $|s \triangle t| = 1$ (they differ in exactly one coordinate).
- *Johnson graph* $J(n, \ell)$: vertices are the ℓ -subsets of $[n]$, and $A \sim B$ iff $|A \cap B| = \ell - 1$. Wikipedia (https://en.wikipedia.org/wiki/Johnson_graph): “two vertices are adjacent when the intersection of the two vertices (subsets) contains” $(k - 1)$ elements.
- *Laplacian* $L = D - A$ (Mathlib’s `SimpleGraph.lapMatrix`).
- *Algebraic connectivity* $a(G) = \lambda_2(L)$. Wikipedia (https://en.wikipedia.org/wiki/Algebraic_connectivity): “The algebraic connectivity (also known as Fiedler value or Fiedler eigenvalue after Miroslav Fiedler) of a graph G is the second-smallest eigenvalue (counting multiple eigenvalues separately) of the Laplacian matrix of G .”
- *Normalized Laplacian* $\mathcal{L} = I - D^{-1/2}AD^{-1/2}$: entry 1 on the diagonal (0 at an isolated vertex), $-1/\sqrt{d_i d_j}$ for adjacent $i \neq j$, and 0 otherwise.

Readings. We refute three readings.

- (J) The appositive “a Johnson graph” fixes the layer graph as $J(n, \ell)$, and $a = \lambda_2(L)$.
- (N) The same graph, with λ_2 of the normalized Laplacian $\mathcal{L}$.
- (I) The literal reading: the subgraph of Q_n induced on the layer, with $a = \lambda_2(L)$.

Since $\ell = n/2$ must be an integer, the order statement is read along even n . $f = \Theta(g)$ means $cg(n) \leq f(n) \leq Cg(n)$ for all large even n , with constants $c, C > 0$. Under (J) and (N) we refute the upper half $f = O(\log n/n^2)$, and under (I) the lower half. Either refutes the conjunction.

Scope. Under (J), every non-trivial layer has $\lambda_2 = n$. So $\ell = n/2$ is a minimiser over the layers $1 \leq \ell \leq n-1$, though not the unique one, and we do not refute the location clause. The conjecture fails through its order clause. Layers 0 and n are single vertices, where λ_2 is undefined. In Lean, `lambda2` is set to 0 when there are fewer than two eigenvalues, and these layers are excluded from the minimum ($1 \leq k \leq n-1$).

2 From a quadratic-form bound to λ_2

In Lean, λ_2 of a real symmetric matrix M indexed by V with $|V| \geq 2$ is `hM.eigenvalues_0` at index $|V| - 2$. Mathlib's list `eigenvalues_0` is sorted in decreasing order (`eigenvalues_0_antitone`), so this is the second-smallest eigenvalue counted with multiplicity.

Lemma 1 (`lambda2_ge`). *Let M be real symmetric with $M\mathbf{1} = 0$, let $c > 0$, and suppose $c \sum_i x_i^2 \leq x^\top Mx$ for every x with $\sum_i x_i = 0$. If $|V| \geq 2$, then $c \leq \lambda_2(M)$.*

Proof. Let (v_j) be Mathlib's orthonormal eigenbasis, $Mv_j = \mu_j v_j$. Since M is symmetric and $M\mathbf{1} = 0$, we have $\mathbf{1}^\top Mx = 0$ for all x , so $\mu_j \sum_t v_j(t) = 0$. If $\mu_j \neq 0$, then $v_j \perp \mathbf{1}$ and $c = c\|v_j\|^2 \leq v_j^\top Mv_j = \mu_j$. So every eigenvalue is 0 or at least c . Moreover, $\mu_j = 0$ forces $\sum_t v_j(t) \neq 0$: otherwise $c \leq v_j^\top Mv_j = 0$. Suppose $i \neq j$ and $\mu_i = \mu_j = 0$. Put $a = \sum v_j$, $b = \sum v_i$ and $x = av_i - bv_j$. Then $\sum x = 0$ and $Mx = 0$, so $c(a^2 + b^2) = c\|x\|^2 \leq 0$, which contradicts $b \neq 0$. Hence at most one index carries an eigenvalue below c . Because the list is sorted in decreasing order, the entries at indices $|V| - 2$ and $|V| - 1$ come from two different indices. So the entry at $|V| - 2$ is at least c . $\square$

3 The Johnson graph: $\lambda_2 \geq n$ for all n

For f defined on subsets of $[n]$, put

$$(Df)(C) = \sum_{i \notin C} f(C \cup \{i\}), \quad (Uf)(E) = \sum_{i \in E} f(E \setminus \{i\}), \quad Xf(A) = \sum_{j \in A} \sum_{i \notin A} f((A \setminus \{j\}) \cup \{i\}).$$

Write $\mathcal{S}_m$ for the family of m -subsets of $[n]$.

Lemma 2. (a) *`up_dn`, `dn_up`: $UDf(A) = |A|f(A) + Xf(A)$ and $DUf(A) = (n - |A|)f(A) + Xf(A)$.* (b) *`adjoint`: $\sum_{E \in \mathcal{S}_{m+1}} g(E)Uh(E) = \sum_{A \in \mathcal{S}_m} h(A)Dg(A)$.* (c) *`sum_dn`: $\sum_{\mathcal{S}_m} Dg = (m+1) \sum_{\mathcal{S}_{m+1}} g$.*

Proof. (a) Expand both double sums; the diagonal term ($i = j$) gives $f(A)$ each time. (b) Both sides sum over pairs (E, i) with $i \in E$, which correspond to pairs (A, i) with $i \notin A$ via $A = E \setminus \{i\}$. (c) Apply (b) with $h \equiv 1$. $\square$

Lemma 3 (key). *If $m+1 \leq n$ and $\sum_{\mathcal{S}_{m+1}} f = 0$, then $\sum_{\mathcal{S}_m} (Df)^2 \leq m(n-m-1) \sum_{\mathcal{S}_{m+1}} f^2$.*

Proof. We induct on m . For $m = 0$, $\mathcal{S}_0 = \{\emptyset\}$ and $Df(\emptyset) = \sum_{\mathcal{S}_1} f = 0$ by Lemma 2(c). For the step, let f live on $\mathcal{S}_{m+2}$ with sum 0, and let $z = Df$, which has sum 0 by (c). Put $Z = \|z\|^2$ and $F = \|f\|^2$. By (b), $Z = \sum_{\mathcal{S}_{m+2}} f Uz$. By (b) twice and (a), $\|Uz\|^2 = \|Dz\|^2 + (n - 2(m+1))Z$. The induction hypothesis then gives $\|Uz\|^2 \leq (m+1)(n-m-2)Z$. Cauchy-Schwarz gives $Z^2 \leq F\|Uz\|^2 \leq (m+1)(n-m-2)FZ$, so $Z \leq (m+1)(n-m-2)F$ (trivially if $Z = 0$). $\square$

Theorem 4 (`johnson_quad`, `algConn_johnson_ge`). *For $k = m+1 \geq 1$ and every x on $\mathcal{S}_k$ with $\sum x = 0$: $n\|x\|^2 \leq x^\top Lx$, where L is the Laplacian of $J(n, k)$. Hence $\lambda_2(L(J(n, k))) \geq n$ for all n and all $1 \leq k \leq n-1$.*

Proof. The neighbours of A in $J(n, k)$ are exactly the sets $(A \setminus \{j\}) \cup \{i\}$ with $j \in A$ and $i \notin A$, and this description is injective (`nbr_sum`). Hence $\deg A = k(n-k)$ (`degree_johnson`) and $\sum_{B \sim A} x_B = Xx(A)$. By Lemma 2(a), $(Lx)(A) = k(n-k+1)x_A - U Dx(A)$, and by (b), $x^\top Lx = k(n-k+1)\|x\|^2 - \|Dx\|^2$. Lemma 3 gives $\|Dx\|^2 \leq m(n-m-1)\|x\|^2$, and $(m+1)(n-m) - m(n-m-1) = n$. Finally, $L\mathbf{1} = 0$ and $|S_k| = \binom{n}{k} \geq 2$, so Lemma 1 applies with $c = n$. $\square$

Remark 5. The vector $x_A = [0 \in A] - k/n$ satisfies $Lx = nx$, so in fact $\lambda_2 = n$. Lean proves only the lower bound, which is all the disproof needs.

Corollary 6 (`normConn_johnson_ge`). *For $1 \leq k \leq n-1$, $\lambda_2(\mathcal{L}(J(n, k))) \geq n/(k(n-k))$; for even n and $k = n/2$ this is $4/n$.*

Proof. $J(n, k)$ is d -regular with $d = k(n-k) > 0$, so $\mathcal{L} = d^{-1}L$ (`normLap_johnson`). Apply Lemma 1 with $c = n/d$, using Theorem 4. $\square$

4 The literal reading

Theorem 7 (`algConn_inducedLayer`). *For all n, ℓ , the subgraph of Q_n induced on the layer of weight ℓ has no edges. Its Laplacian is 0, and its λ_2 is 0.*

Proof. If $|s| = |t| = \ell$, then $|s \setminus t| = |t \setminus s|$, so $|s \triangle t| = 2|s \setminus t| \neq 1$ (`inducedLayer_no_adj`). Every eigenvalue of the zero matrix is 0 (`lambda2_zero`). $\square$

5 Asymptotics and the main theorem

Lemma 8 (`exists_large`). *For every $C \in \mathbb{R}$ and $N \in \mathbb{N}$, there is an even $n \geq \max(N, 2)$ with $C \log n < 4n$.*

Proof. Take $n = 2(N + M + 1)$ with $M \geq C^2$. Since $\log \sqrt{n} \leq \sqrt{n} - 1$, we get $\log n < 2\sqrt{n}$. If $C \leq 0$, the claim is clear. Otherwise $C \leq \sqrt{n}$, so $C \log n < 2\sqrt{n}\sqrt{n} = 2n < 4n$. $\square$

Theorem 9 (`conjecture_3963_false`). *All four of the following statements are false:*

1. (*J, minimum over layers; johnson_min_not_bigO*) there are C, N such that for every even $n \geq N$ some layer $1 \leq k \leq n-1$ has $\lambda_2(L(J(n, k))) \leq C \log n/n^2$;
2. (*J, middle layer; johnson_not_bigO*) there are C, N with $\lambda_2(L(J(n, n/2))) \leq C \log n/n^2$ for all even $n \geq N$;
3. (*N, middle layer; johnson_norm_not_bigO*) the same bound for $\lambda_2(\mathcal{L}(J(n, n/2)))$;
4. (*I, middle layer; induced_not_bigOmega*) there are $c > 0$ and N with $c \log n/n^2 \leq \lambda_2$ of the induced layer graph of weight $n/2$, for all even $n \geq N$.

Proof. (1)–(2): By Theorem 4, every non-trivial layer has $\lambda_2 \geq n$. With n from the lemma, $C \log n < 4n \leq n^3$, so $C \log n/n^2 < n$. (3): For $n = 2t$, $\lambda_2 \geq 2t/t^2$, and $C \log n \cdot t^2 < 8t \cdot t^2$ gives $C \log n/n^2 < 2t/t^2$. (4): $\lambda_2 = 0 < c \log n/n^2$ for $n \geq 2$. $\square$

Hence the conjectured order $\Theta(\log n/n^2)$ fails under readings (J), (N) and (I). Under (J) and (N) the true order is n and $4/n$ respectively; under (I) the value is identically 0.

6 Lean formalization

The file `lean/Conjecture3963/Basic.lean` (namespace `C3963`, only import `Mathlib`, 564 lines) defines `hypercube`, `Layer`, `johnson`, `lambda2`, `algConn` (`lambda2` of `lapMatrix`), `normLap` and `inducedLayer`. It proves every statement above under the names given. The main theorem is `C3963.conjecture_3963_false`; the all- n spectral bound is `C3963.algConn_johnson_ge`. `#print` axioms for both gives exactly `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.